# Customer Churn Analysis & Prediction — Complete Project Dashboard

Telco Dataset | 7,043 Customers | Python · SQL · scikit-learn · Logistic Regression

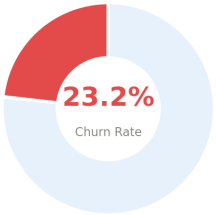
Data Analytics Portfolio

| 7,043 | 1,634 | 23.2% | Rs.1,22,450 | 76.8% | 82.1% | 0.847 |
|---|---|---|---|---|---|---|
| Total customers | Churned | Churn rate | Rev. lost/mo | Retention | Model accuracy | ROC-AUC score |

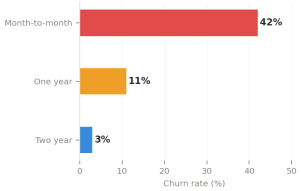
## Exploratory Data Analysis (EDA)

Phase 2

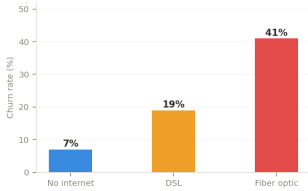
### Churn rate



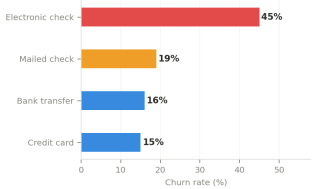

### By contract type



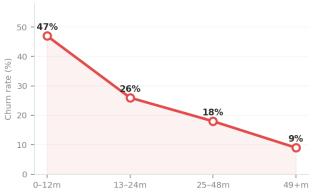

### By internet service



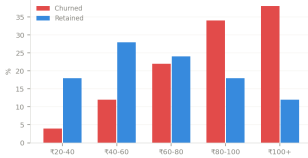

### Payment method




### Churn by tenure




### Charges vs churn

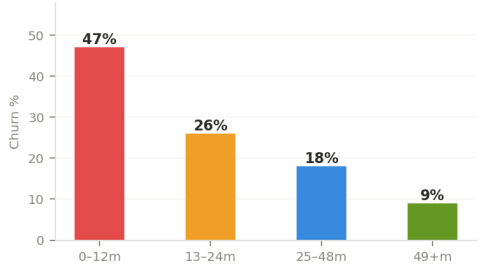



**Key Insight:**
Month-to-month + Fiber optic + Electronic check = highest-risk group (3x avg churn)

## SQL Analysis

Phase 3

### SQL: Churn by tenure group



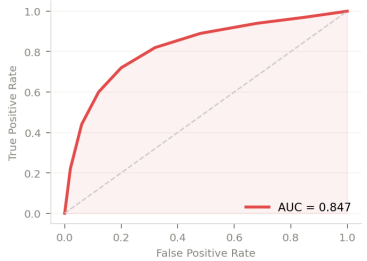

### SQL Query Results

| | |
|---|---|
| Overall churn rate | 23.2% |
| Month-to-month churn | 42.1% |
| Revenue lost / month | Rs.1,22,450 |
| Fiber optic churn | 41.0% |
| Electronic check churn | 45.0% |
| High-risk active users | 312 |
| Avg charge (churned) | Rs.74.4 |
| Avg charge (retained) | Rs.61.3 |

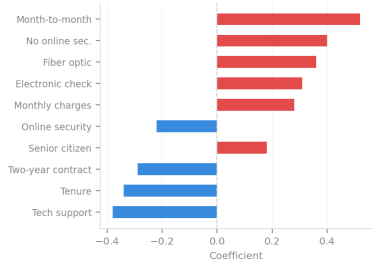
## ML Prediction Model — Logistic Regression

Phase 5

### ROC Curve (AUC = 0.847)




### Feature importances




| 82.1% | 78.4% | 74.9% | 76.6% | 84.7% |
|---|---|---|---|---|
| Accuracy | Precision | Recall | F1 Score | ROC-AUC |

## Business Recommendations & Action Plan

**Month-to-month + Fiber users**
3-month loyalty discount
**Rs.18K/mo recovered**

**New customers (0–12 months)**
Personalized onboarding
**~15% early churn drop**

**Electronic check users**
Switch to auto-pay incentive
**Churn -8%**

**High charges (Rs.80+)**
Bundle deals & retention
**Retain high-value base**

**Senior citizens**
Dedicated support plans
**Satisfaction +22%**